

Laboratory Report: Exercise 2

Computer Networks Laboratory

Introduction to GNS3 and Basic Network Configuration

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ΠΑΝΕΠΙΣΤΗΜΙΟ
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1. Main Topology & Script Success

This screenshot displays the complete network topology in GNS3 and the successful execution of the evaluation script. The connectivity tests confirm **GW_SUCCESS**, **8.8.8.8_SUCCESS**, and **google.com SUCCESS**, proving that the routing protocols and the DNS configuration are working correctly across the entire infrastructure.

```
debian@debian:~$ sudo ./2nd_Ergasia.sh.x
Έλεγχος εγκαταστημένων προγραμμάτων:
Τα πρόγραμματα είναι σωστά εγκατεστημένα.
Give your Arithmo Mitroou/Δώστε τον αριθμο μητρώου (7ψήφιο) σας: 1125544
your ip address: 112.55.44.2 and your gateway: 112.55.44.1
Έύρεση εγγραφής στο eclass....
Δώστε όνομα/επίθετο:Alvaro Gonzalez Riquelme
Δώστε το email που έχετε δηλωμένο στο eclass:up1125544@ac.upatras.gr
ok
Checking Gateway: GW_SUCCESS
begin ping to gateway: ping to GW_SUCCESS
begin ping to 8.8.8.8: ping to 8.8.8.8_SUCCESS
Begin ping to google.com: ping to google.com: SUCCESS
Start traceroute to 8.8.8.8 with ICMP: Αδυναμία συνέχισης. error 214. Επικοινωνήστε με τον διδάσκοντα.
```

2. NAT Translations Table

The output of the show ip nat translations command on **R1** verifies the active Port Address Translation (PAT). It shows how the internal private traffic is being mapped to the gateway's public IP address (192.168.122.152), enabling the Debian client to access external internet resources through the NAT cloud.

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 192.168.122.152:24212 10.0.12.2:24212 8.8.8.8:24212      8.8.8.8:24212
```

3. Routing Table on R2

This capture shows the routing table of **R2**. The presence of the static default route (0.0.0.0/0) pointing towards R1's interface ensures that all traffic from the Debian client is correctly forwarded to the gateway, while the local connected routes maintain communication with the internal subnet.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.12.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.12.1
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C      10.0.12.0/24 is directly connected, Ethernet1/0
L      10.0.12.2/32 is directly connected, Ethernet1/0
O      10.0.13.0/24 [110/11] via 10.0.12.1, 00:19:56, Ethernet1/0
C      10.0.23.0/24 is directly connected, Ethernet1/1
L      10.0.23.2/32 is directly connected, Ethernet1/1
      112.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      112.55.44.0/24 is directly connected, Ethernet1/2
L      112.55.44.1/32 is directly connected, Ethernet1/2
--More--
```

4. Interface Status on R1

This final capture shows the output of show ip interface brief on the gateway router. It confirms that all critical interfaces are in an **'up/up'** state. Specifically, it highlights that **FastEthernet 0/0** has successfully obtained its IP address via DHCP from the NAT cloud, and the internal interfaces connecting to the other routers are correctly addressed and operational.

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.122.152 YES DHCP    up          up
FastEthernet1/0          10.0.12.1       YES NVRAM   up          up
FastEthernet1/1          10.0.13.1       YES NVRAM   up          up
NVI0                     unassigned      YES unset  administratively down down
R1#
```